

KASRA SIAVASHPOUR

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EDUCATION

Sharif University of Technology (SUT) , Tehran, Iran	2020–Jul 2026
Bachelor of Science, Computer Science, Department of Mathematical Sciences	
Allameh Tabatabaiei High School (selective, non-governmental) , Tehran, Iran	2016–2020
High School Diploma, Mathematics and Physics	

RESEARCH INTERESTS

- Distributed & Decentralized Systems
- IoT & Cyber-Physical Systems
- Multi-Agent Systems
- AI-Driven Network Automation

MANUSCRIPTS

Submitted

- A. Vaezi, S. Niknejad, **K. Siavashpour**, A. Dadras, “FAIR: A Novel Envy-Free Multi-Agent Model for Traffic Systems,” submitted to the **International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2027**.

In Preparation

- A. Siavashpour, **K. Siavashpour**, M. Azadegan, B. Safaei, H. Peyvandi, M. Imran, “A Comprehensive Survey of Self-Organizing Networks in Next-Generation Cellular Systems,” to be submitted to the **IEEE Open Journal of the Communications Society (OJ-COMS)**.
- **K. Siavashpour**, with A. Vaezi, A. Daneshgar and A. Dadras (author order not final), “The Loop of the Rings: A Fully Decentralized Reliable Cooperative System,” manuscript in preparation for submission to the **IEEE Symposium on Foundations of Computer Science (FOCS) 2027**. [[arXiv preprint](#)]

RESEARCH EXPERIENCE

Research Assistant, Institute for Research in Fundamental Sciences (IPM)

& **Sharif University of Technology**, Tehran, Iran

May 2026–present

Supervisor: [Dr. Arash Vaezi](#)

Project: “FAIR: A Novel Envy-Free Multi-Agent Model for Traffic Systems”

- Built the discrete-event simulator and designed the evaluation: 7 OpenStreetMap city networks (9K–20K edges, 33K–132K origin–destination trips each), same parameters in every city.
- My simulations exposed counterexamples to two lemmas of an earlier draft; we restated them with Dr. Vaezi.
- Against a congestion-oblivious shortest-path baseline, FAIR lowered envy spread by about an order of magnitude and nearly eliminated trips taking over twice their predicted time.

Research Assistant, Institute for Research in Fundamental Sciences (IPM)

& **Sharif University of Technology**, Tehran, Iran

Aug 2024–present

Supervisors: [Dr. Arash Vaezi](#), [Dr. Amir Daneshgar](#)

Project: “The Loop of the Rings: A Fully Decentralized Reliable Cooperative System”

- Co-developed LoR’s Go simulator with A. Dadras: cooperation rings, fractal rings, verification teams and the trader-behaviour model.

- Designed and ran the simulation study on 500-trader instances, sweeping the shares of malicious and erratic traders.
- With at least 60% honest traders, verification accuracy stayed above 90% and satisfaction above 80%; satisfaction fell below 20% once dishonest traders passed 60%.

Research Assistant, Sharif University of Technology, Tehran, Iran

Aug 2024–present

Supervisors: [Dr. Hossein Peyvandi](#), [Dr. Masoumeh Azadegan](#) (Khatam University)

Collaborators: [Dr. Bardia Safaei](#), [Dr. Muhammad Imran](#) (University of Glasgow)

Project: “A Comprehensive Survey of Self-Organizing Networks in Next-Generation Cellular Systems”

- Co-wrote the survey with A. Siavashpour in equal shares, covering 114 references; the other authors advised and reviewed.
- Classified 106 papers by SON function and period: self-optimization accounts for 61%, and self-healing grew from a single paper (2014–16) to 24.
- Identified open problems, including conflicts between SON functions and SON’s role in 6G.

Research Assistant, Politecnico di Milano, Milan, Italy (Remote)

May 2024–Aug 2024

Supervisor: [Dr. Gianni Antichi](#)

- Researched WebAssembly for systems acceleration and designed a benchmarking plan for CPU, memory, latency and throughput.

Research Assistant, Sharif University of Technology, Tehran, Iran

Oct 2023–Jan 2024

Supervisor: [Dr. Bardia Safaei](#)

- Simulated a decentralized flying ad hoc network (FANET) of UAVs in NS-3 with ant-colony-based cluster routing.
- Implemented beaconing and neighbor discovery under UAV mobility, and analyzed energy–delivery trade-offs.
- Built reproducible pipelines (configs, seeds, scripts) for experiment reruns.

NOTABLE PROJECTS

SliceDB: Distributed Key-Value Store (*Go, etcd*) [[GitHub Link](#)]

2025, extended 2026

Course project with A. Siavashpour

- Hash-partitioned key-value store with write-ahead-log replication from leaders to followers, etcd leases for membership and a fenced controller election.
- Online resharding under load (8 to 16 partitions); leader handoff loses no acknowledged write.
- `kill -9` of a node under load: 0 client-visible errors (16 retries), writes paused about 4s.

Multimodal Product Search (*Python, CLIP, FAISS*) [[GitHub Link](#)]

2025–2026

Course project, Machine Learning Systems Design

- Text and image retrieval over Torob’s catalog with multilingual sentence embeddings, CLIP and FAISS (text index of 1.02M products, image index of 100K), tuned with Optuna and evaluated on 5,000 logged user queries.
- Added a chat agent that uses retrieval as a tool and keeps per-conversation memory.

Highly Available HDFS + Spark (*Hadoop, ZooKeeper, Spark*) [[GitHub Link](#)]

2025

Course project with A. Siavashpour

- Deployed HDFS with automatic NameNode failover and ran Spark queries on NYC taxi data.

Go Judge: Online Judge (*Go, PostgreSQL, Docker*) [[GitHub Link](#)]

2025

Course project with A. Siavashpour

- Built the web server, UI and database schema; each submission runs in a Docker sandbox.

TEACHING EXPERIENCE

Teaching Assistant, Sharif University of Technology

- **Computer Networks** — Dr. Mahdi Dolati Fall 2026, Winter 2026
- **Operating Systems** — Dr. Hossein Asadi Fall 2026
- **Operating Systems** — Dr. Hossein Asadi & Dr. Rasul Jalili Fall 2024
- **Computer Networks** — Dr. Hossein Peyvandi Winter 2024
- **Operating Systems** — Dr. Arash Vaezi [\[Piazza\]](#) Winter 2024
- **Computer Networks** — Dr. Kambiz Mizanian Fall 2023
- **Operating Systems** — Dr. Mohaddeseh Mirbeygi & Dr. MohammadAli Mirzaei Fall 2023

Designed and graded weekly assignments (120+ students in the largest classes), wrote rubrics, held office hours, guided projects and built autograders.

HONORS & AWARDS

- Top 0.1% among 155,250 candidates, Iranian University Entrance Exam (B.Sc.) 2020
- 1st place in National Hovercraft Competition (high school) — design & control 2016–2017

SKILLS

- Programming Languages:** Go, Python, Java, C/C++, Bash

Libraries: PyTorch, scikit-learn, FAISS, NumPy, Matplotlib
- Systems & Tools:** etcd, Docker, PostgreSQL, Spark, NS-3, Git

Languages: Persian (Native), English (C1)